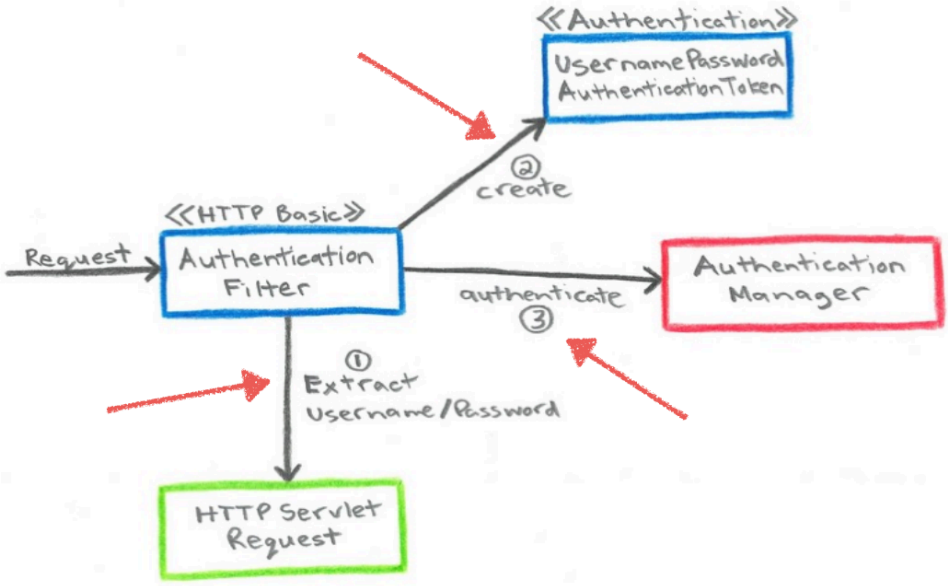
Spring-security - Intro

Authentication

Authorization

Exception Handling

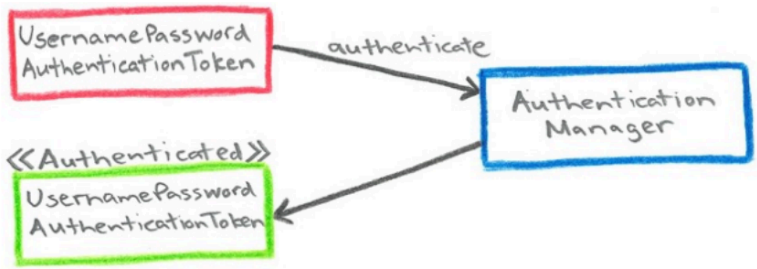
Authentication Filter



Authentication

Authentication	
Principal:	joe@example.com
Credentials:	password
Authorities:	——
Authenticated:	FALSE

Authentication	
Principal:	UserDetails
Credentials:	——
Authorities:	ROLE_USER
Authenticated:	TRUE




```
public interface Authentication extends Principal, Serializable {

   Object getPrincipal();

   Object getCredentials();

   Collection<? extends GrantedAuthority> getAuthorities();

   . . .
}
```

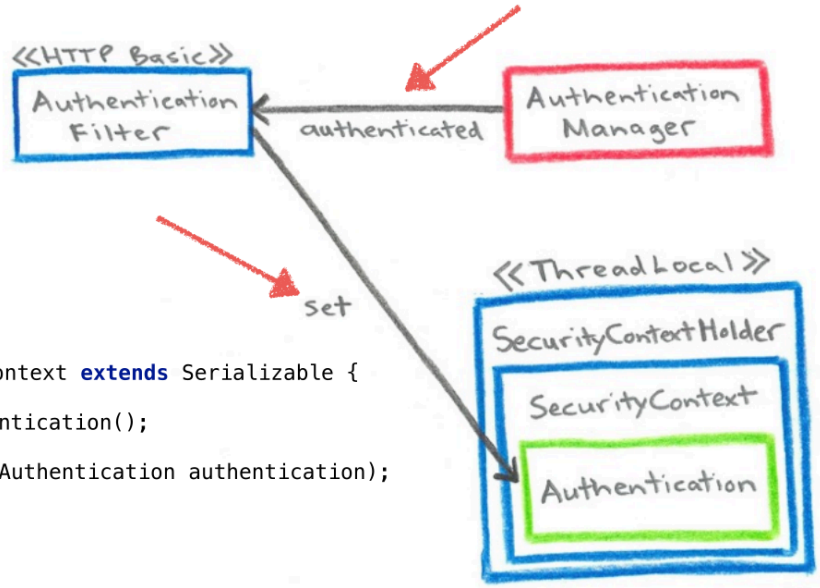
UserDetails / Service

```
public interface UserDetailsService {

   UserDetails loadUserByUsername(String username)
         throws UsernameNotFoundException;

}
```

Authentication	
Principal:	UserDetails
Credentials:	——
Authorities:	ROLE_USER
Authenticated:	TRUE

```
public interface UserDetails extends Serializable {

   String getUsername();

   String getPassword();

   Collection<? extends GrantedAuthority> getAuthorities();

   . . .
}
```

Security Context




```
public interface SecurityContext extends Serializable {
    Authentication getAuthentication();
    void setAuthentication(Authentication authentication);
}
```

```
SecurityContextHolder.getContext().setAuthentication(authenticated);
```

Authentication Recap

- **Authentication Filter** creates an *"Authentication Request"* and passes it to the **Authentication Manager**
- Authentication Manager delegates to the **Authentication Provider**
- Authentication Provider uses a **UserDetailsService** to load the **UserDetails** and returns an *"Authenticated Principal"*
- Authentication Filter sets the **Authentication** in the **SecurityContext**

Authentication

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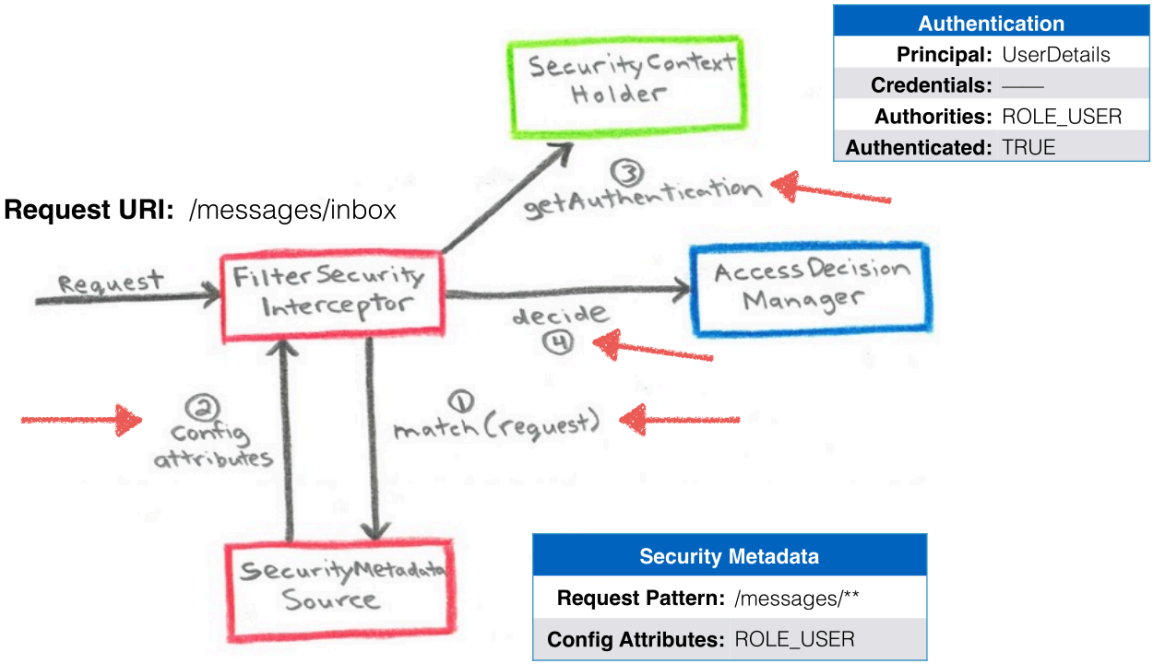
Exception Handling

Filter Security Interceptor

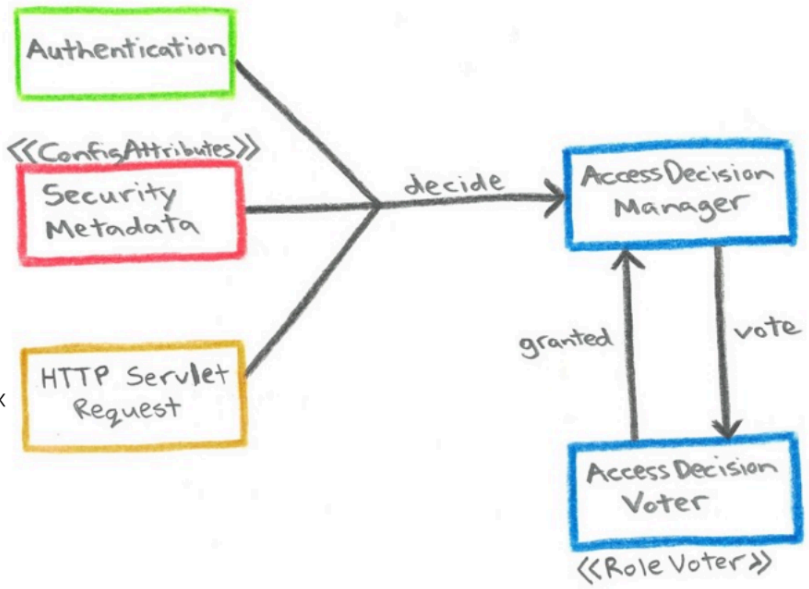


Access Decision

Authentication	
Principal:	UserDetails
Credentials:	——
Authorities:	ROLE_USER
Authenticated:	TRUE

Security Metadata	
Request Pattern:	/messages/**
Config Attributes:	ROLE_USER

Request URI: /messages/inbox



Authorization Recap

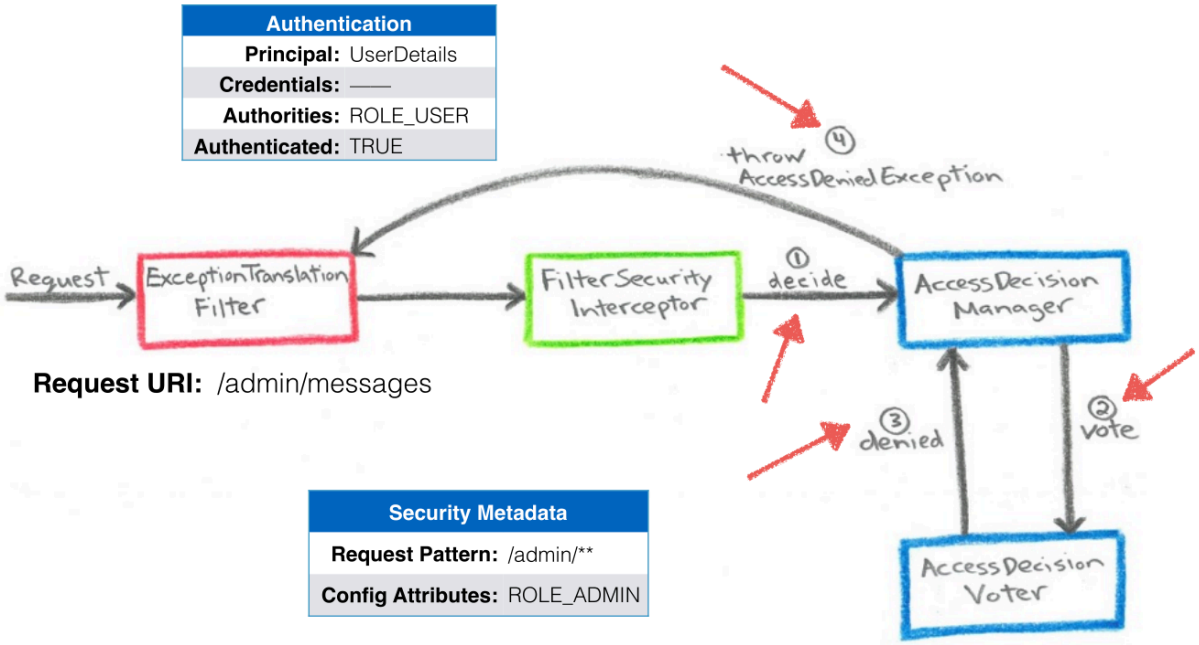
- **FilterSecurityInterceptor** obtains the ***"Security Metadata"*** by matching on the current request
- FilterSecurityInterceptor gets the current **Authentication**
- The Authentication, Security Metadata and Request is passed to the **AccessDecisionManager**
- The AccessDecisionManager delegates to it's **AccessDecisionVoter(s)** for decisioning

Authentication

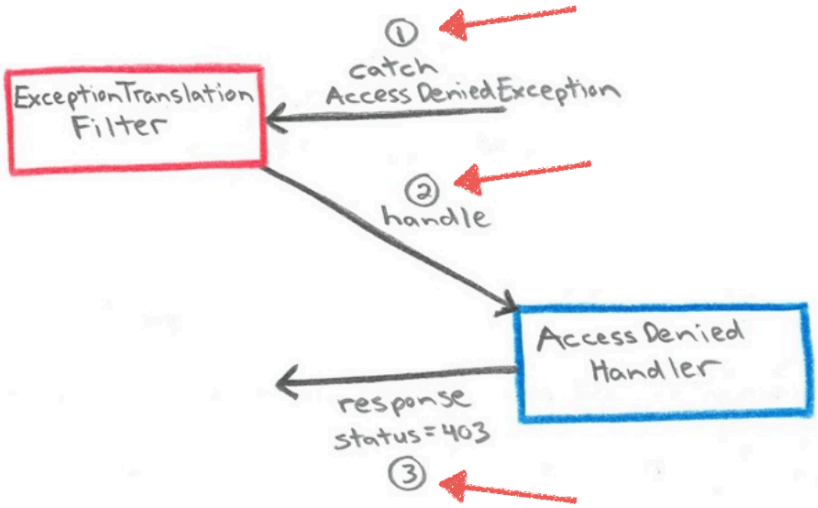
Authorization

Exception Handling

Access Denied

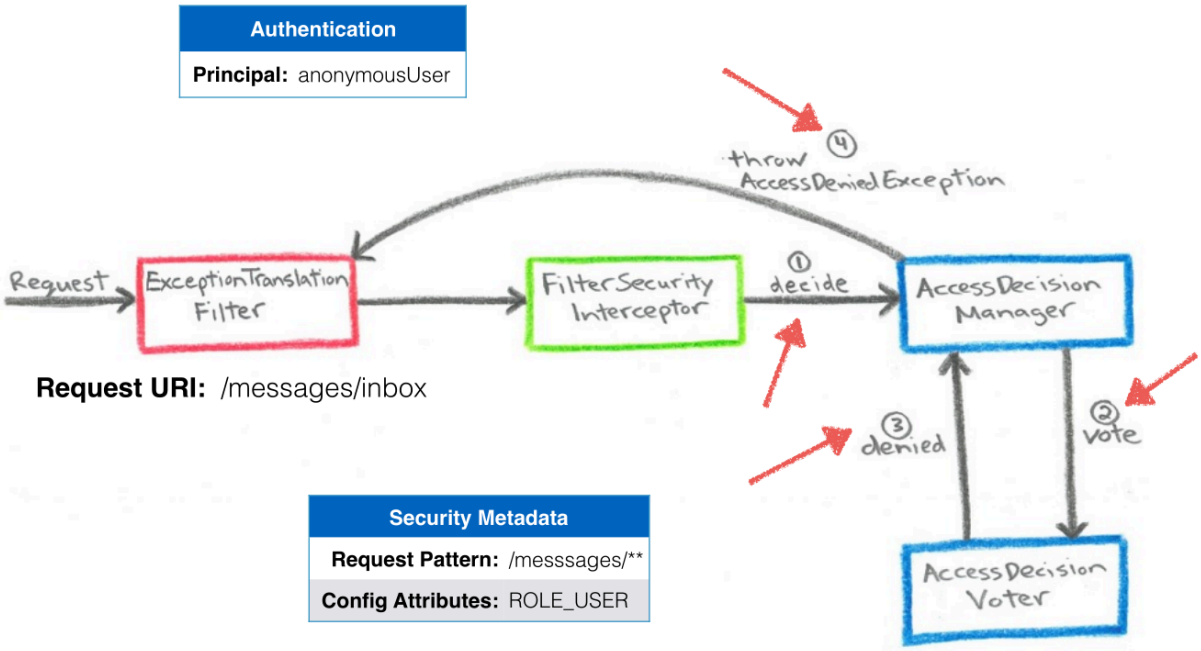


Access Denied Handler

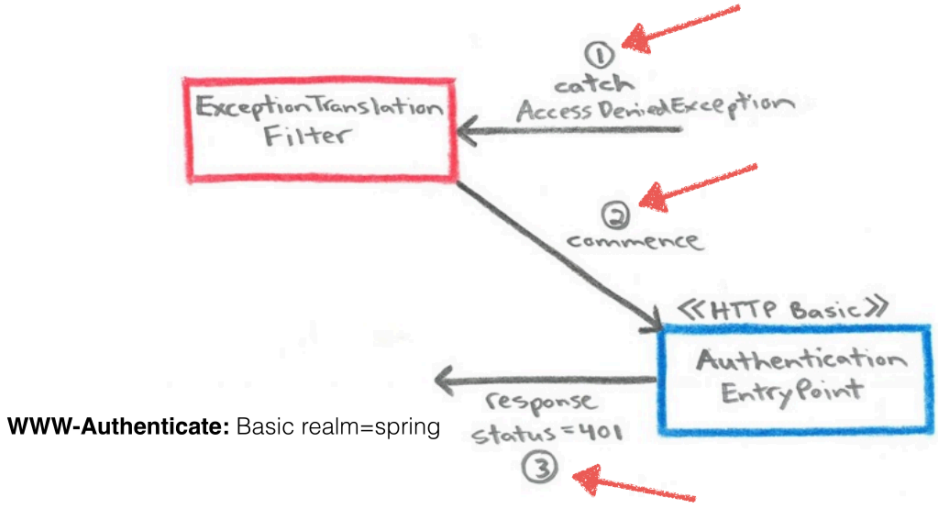



```
public interface AccessDeniedHandler {

  void handle(HttpServletRequest request, HttpServletResponse response,
       AccessDeniedException accessDeniedException) throws IOException, ServletException;

}
```

“Unauthenticated”



Start Authentication




```
public interface AuthenticationEntryPoint {

    void commence(HttpServletRequest request, HttpServletResponse response,
            AuthenticationException authException) throws IOException, ServletException;

}
```

Exception Handling Recap

- When *"Access Denied"* for current Authentication, the **ExceptionTranslationFilter** delegates to the **AccessDeniedHandler**, which by default, returns a 403 Status.

- When current Authentication is *"Anonymous"*, the **ExceptionTranslationFilter** delegates to the **AuthenticationEntryPoint** to start the Authentication process.

https://dzone.com/articles/cookies-vs-tokens-the-definitive-guide